DATA VISUALIZATION DASHBOARD PROJECT

Project Description Document

1. PROJECT OVERVIEW

This project requires each student to build a fully functional, professional-grade data visualization dashboard. The dashboard will analyze a shared dataset and present meaningful insights through multiple chart types, interactive filters, and a clean frontend interface.

This is an **individual project .** You can access your dataset using your SAP ID. If any student is unable to access their SAP ID, or if their SAP ID is not mentioned in the sheet, they must contact their CR or GR within one day of the assigned date. The CR or GR should then share the list of such students with me whose SAP IDs are not mentioned in the sheet. If any student finds that their dataset contains fewer features, they may download a similar type of dataset from Kaggle or any other relevant data-related website. . Each student will work with the same shared dataset but is expected to produce a unique, well-designed dashboard that demonstrates their understanding of data analysis and visualization.

2. OBJECTIVES

• Load, clean, and analyze data using the **Pandas** library

• Create a variety of professional visualizations using **Matplotlib** and **Seaborn**

• Build an interactive frontend dashboard with filters and controls

• Present data insights in a clear, professional, and user-friendly manner

• Use tools such as **Streamlit, Gradio** or similar frameworks to deploy dashboards

3. DATASET GUIDELINES

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• The dataset file name is **standardized**. Students must use the **EXACT same file name** as provided. Do NOT rename the file.

• Students who do not have access should contact the instructor

• The dataset contains many features. Students are encouraged to explore **ALL features** in their analysis.

• Future versions of the dataset may include additional features — build your dashboard to be flexible and scalable.

4. TECHNICAL STACK (REQUIRED TOOLS & LIBRARIES)

Python 3.x	Core Language	Base programming language
Pandas	Backend	Data loading, cleaning, filtering, aggregation
NumPy	Backend	Numerical operations
Matplotlib	Visualization	Core plotting and chart creation
Seaborn	Visualization	Statistical visualizations and styling
Streamlit / Gradio / HTML+JS	Frontend	Interactive dashboard interface (choose ONE)

5. REQUIRED CHART TYPES

1	Pie Chart	Show proportional distribution of a category
2	Histogram	Display frequency distribution of a numerical column
3	Line Chart	Show trends over time or sequence
4	Bar Chart	Compare values across categories
5	Scatter Plot	Relationship between two numerical variables
6	Box Plot	Data spread, median, and outliers
7	Heatmap	Visualize correlation matrix of features
8	Area Chart	Cumulative trends over time
9	Count Plot	Frequency count of categorical variables
10	Violin Plot	Distribution and probability density

Bonus (Optional): Pair Plot, Bubble Chart, Funnel Chart

Each chart must have: a clear **title**, labeled **axes**, a **legend** where applicable, and a consistent professional **color scheme**.

6. DASHBOARD FILTER REQUIREMENTS

- **Date / Time Range Filter** — Filter data by a date range (if applicable)
- **Category Filter** — Dropdown to filter by one or more categories
- **Numerical Range Slider** — Filter by numerical value ranges
- **Multi-Select Filter** — Select multiple values from a column
- **Search / Text Filter** — Filter rows by keyword
- **Reset / Clear Filters** — Button to reset all filters to default

Filters must be connected to ALL charts. When a filter is applied, all charts in the dashboard must update dynamically and simultaneously.

7. DASHBOARD DESIGN STANDARDS

- Use a clean, consistent color theme throughout the dashboard
- Group related charts logically on the same page or section
- Include a **Dashboard Title** and brief description at the top
- Add a sidebar or top navigation bar for filters
- Show **KPI summary cards** at the top: Total Records, Key Averages, Notable Highs/Lows
- Charts must not be cluttered or overlapping
- Use proper font sizes for readability
- Dashboard must be responsive and properly laid out

8. PROJECT FOLDER STRUCTURE

```
/dashboard_project/
■■■ data/
■ ■■■ [dataset_filename.csv] ← Use the EXACT provided file name
■■■ notebooks/
■ ■■■ analysis.ipynb ← Exploratory Data Analysis (EDA)
■■■ app.py ← Main dashboard application file
■■■ charts.py ← Chart / visualization functions
■■■ filters.py ← Filter / data processing functions
■■■ requirements.txt ← All required Python packages
■■■ README.md ← Project documentation
```

9. DELIVERABLES

- Complete project source code (zipped folder)
- A working dashboard (run locally or deployed)
- A README.md explaining how to install dependencies, run the dashboard, and brief insights
- A **5–10 minute demo / presentation** of the dashboard

10. EVALUATION CRITERIA

Correct use of dataset (no renaming)	10
Data cleaning and preprocessing (Pandas)	15
Variety and correctness of charts	25
Chart quality (titles, labels, colors)	10
Filter functionality (interactive & linked)	20

Dashboard design and professionalism	10
Code quality and project structure	5
README and documentation	5
TOTAL	**100**

11. IMPORTANT RULES & NOTES

■ **IMPORTANT — Please read carefully:**

- **DO NOT rename** the dataset file.
- **DO NOT download** a different dataset. Contact your instructor if you cannot access the file.
- All 40 students use the **SAME dataset**. Identical code will be treated as plagiarism.
- Your dashboard design, chart selection, and insights must be **YOUR OWN**.
- Late submissions will be penalized as per course policy.
- You may use additional Python libraries, but **Pandas, Matplotlib, and Seaborn are MANDATORY**.

12. SUBMISSION DETAILS

Submission Date	: 05-June-2026
Submission Mode	: Portal Submission
Instructor Name	: Ali Hassan Sherazi
Course Name	:Exploratory Data Analysis